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Hinata et al.

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(54) **DEVICE FOR PRODUCING OPTICAL CONNECTOR AND METHOD FOR PRODUCING OPTICAL CONNECTOR**

(58) **Field of Classification Search**
CPC G02B 6/3861; G02B 6/385
See application file for complete search history.

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(56) **References Cited**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 46 days.

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(22) Filed: **Mar. 8, 2023**

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(30) **Foreign Application Priority Data**

Mar. 10, 2022 (JP) 2022-037185

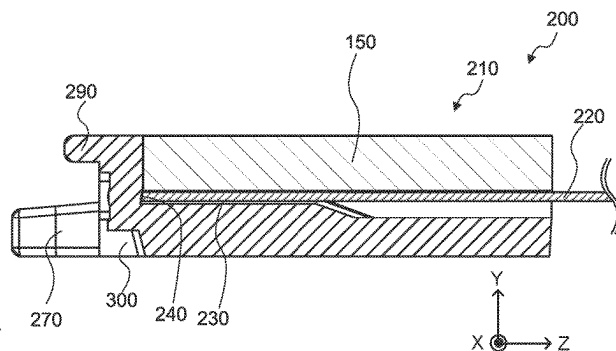
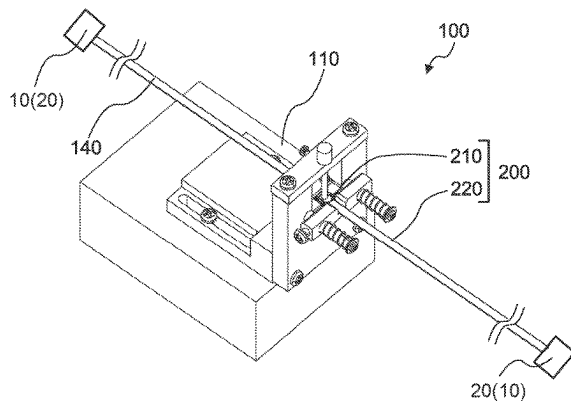
(57) **ABSTRACT**

An optical connector production device of the present invention is configured to produce an optical connector that includes a ferrule and a first optical transmission member. The production device includes a jig body, and a fixing part which allows the ferrule to fit thereto to hold the ferrule and which is disposed in the jig body.

(51) **Int. Cl.**
G02B 6/38 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 6/3861** (2013.01); **G02B 6/385** (2013.01)

4 Claims, 9 Drawing Sheets



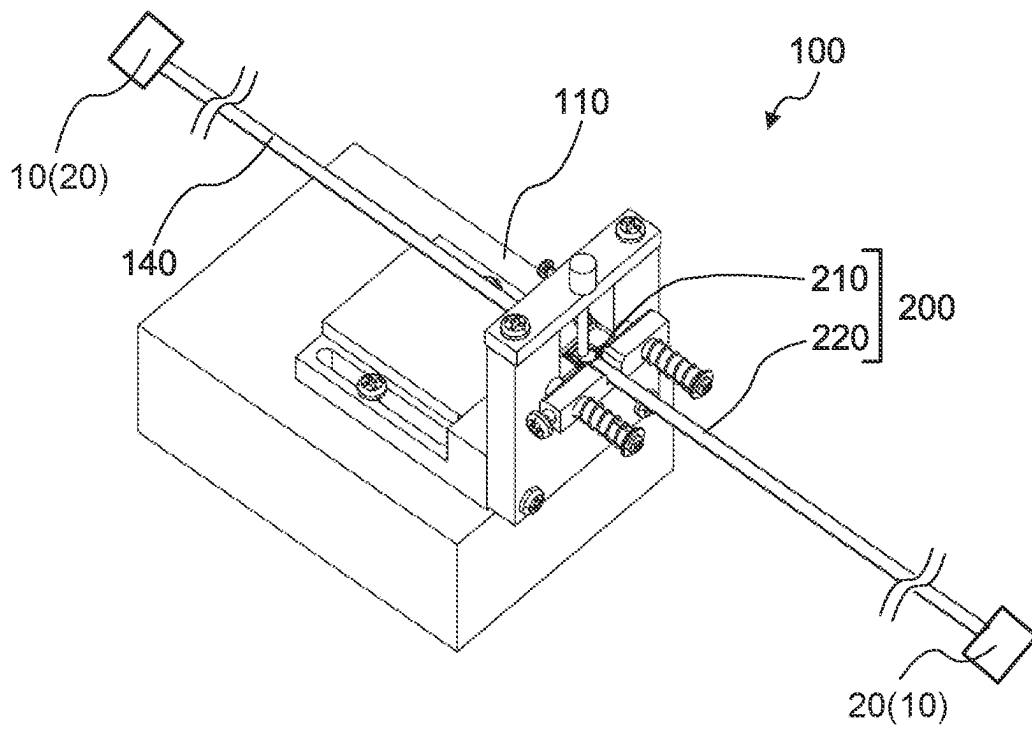


FIG. 1A

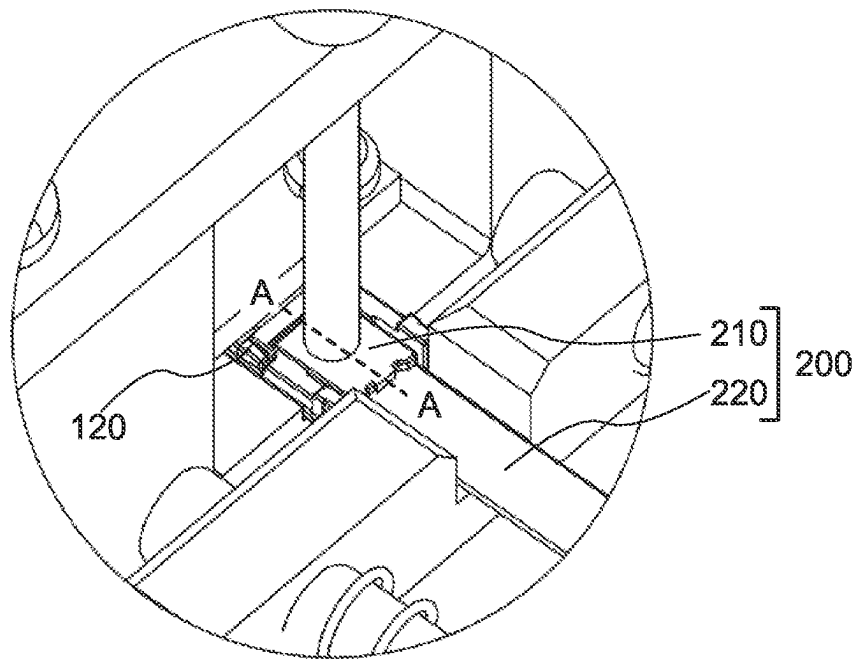


FIG. 1B

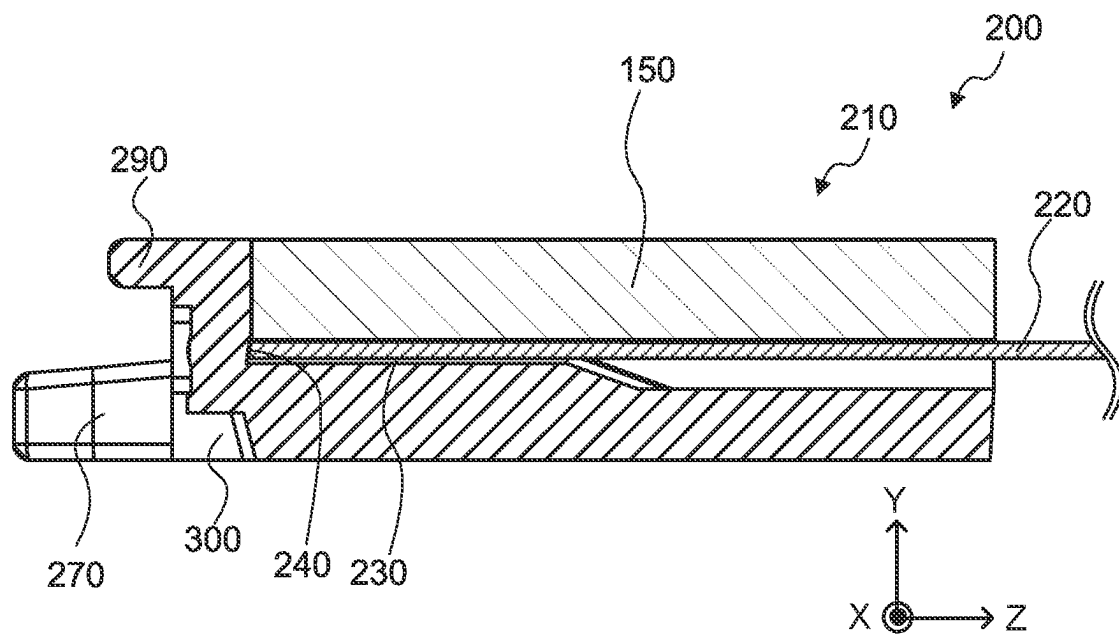


FIG. 2A

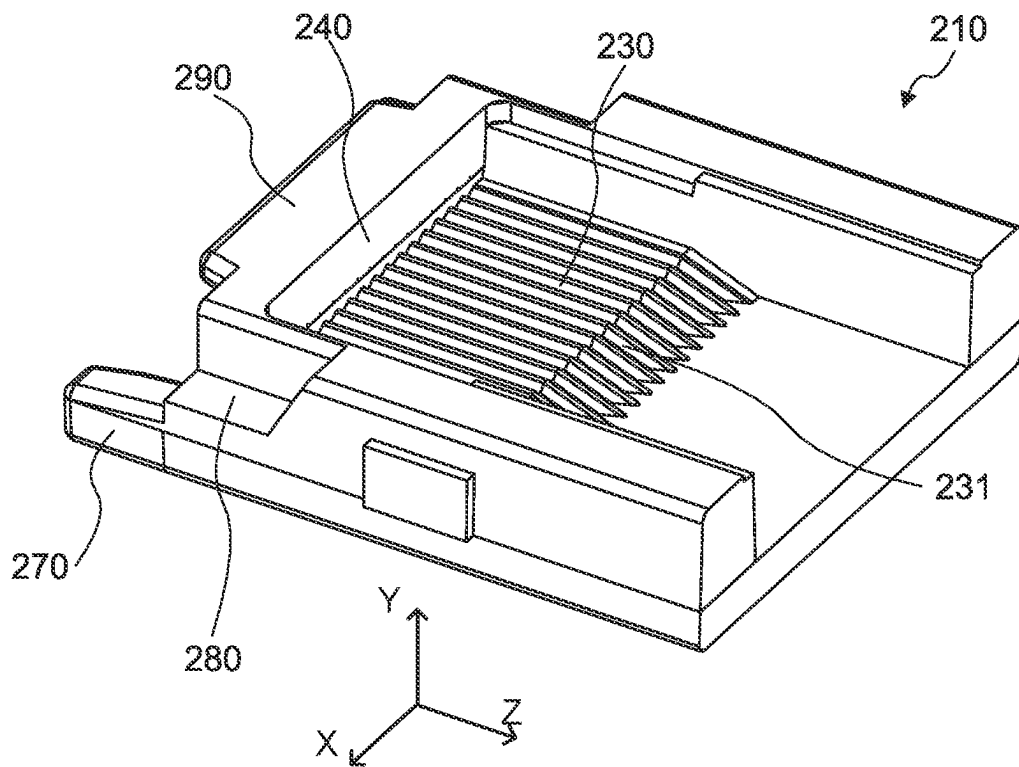


FIG. 2B

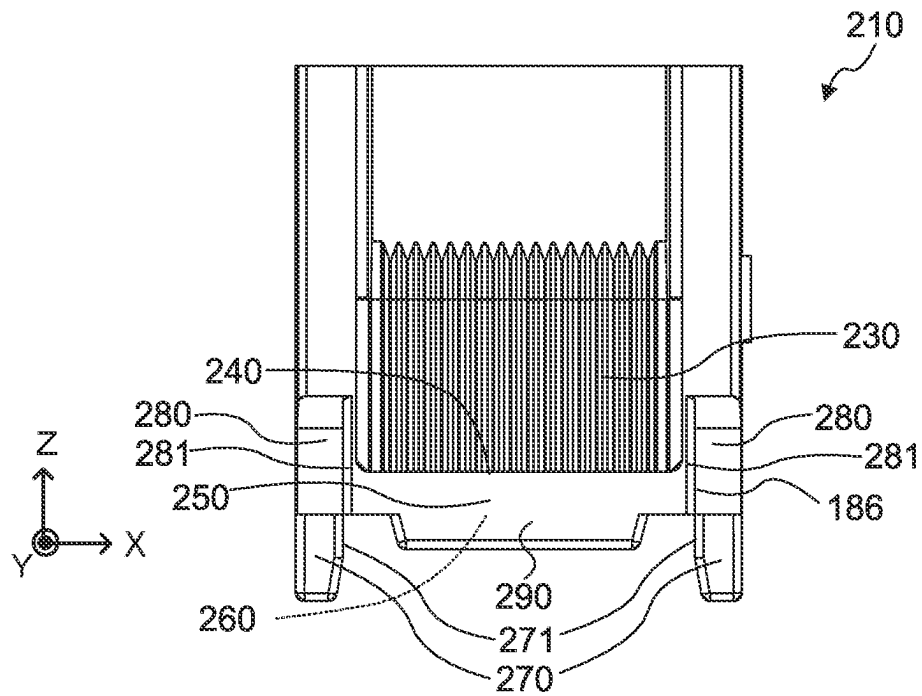


FIG. 3A

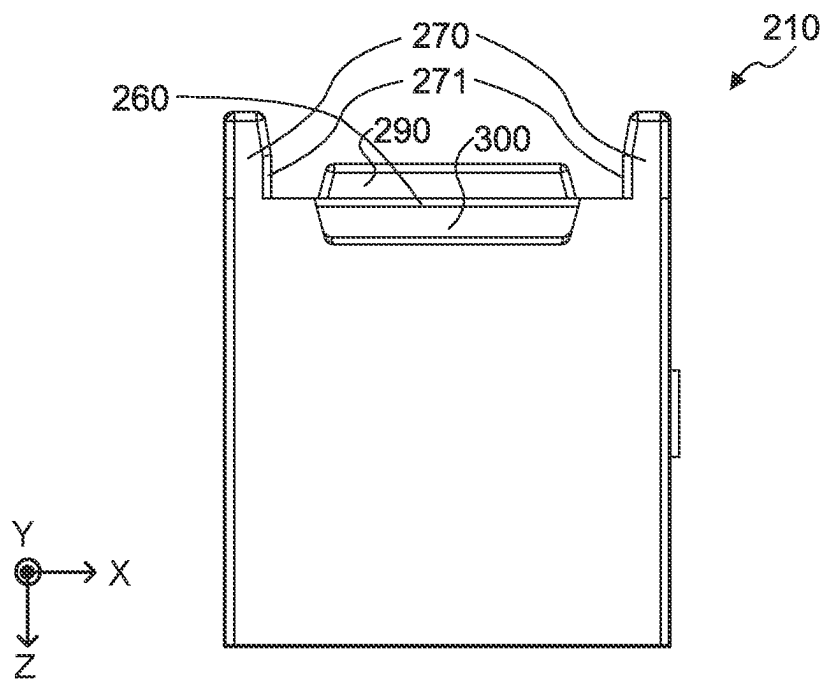


FIG. 3B

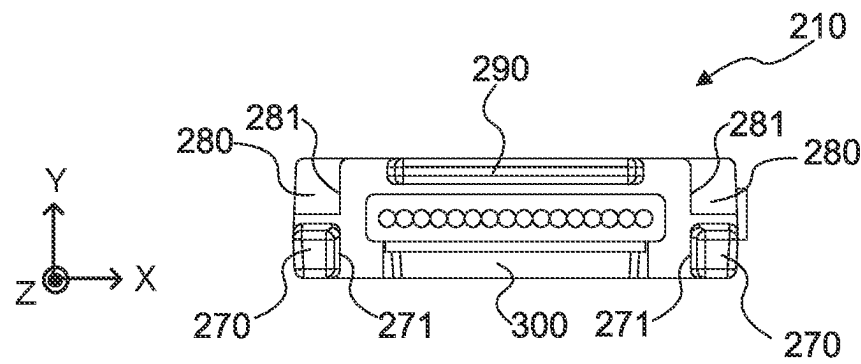


FIG. 4A

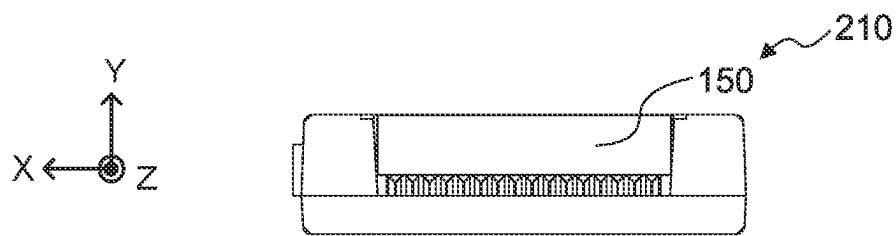


FIG. 4B

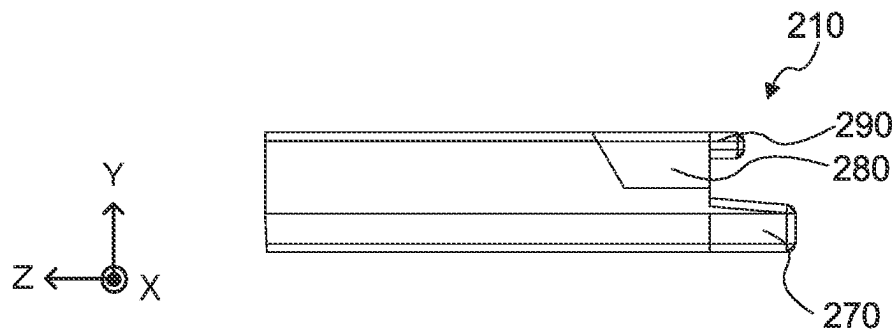


FIG. 4C

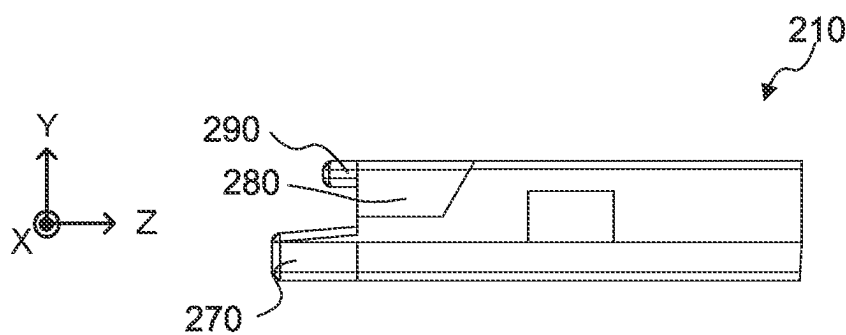


FIG. 4D

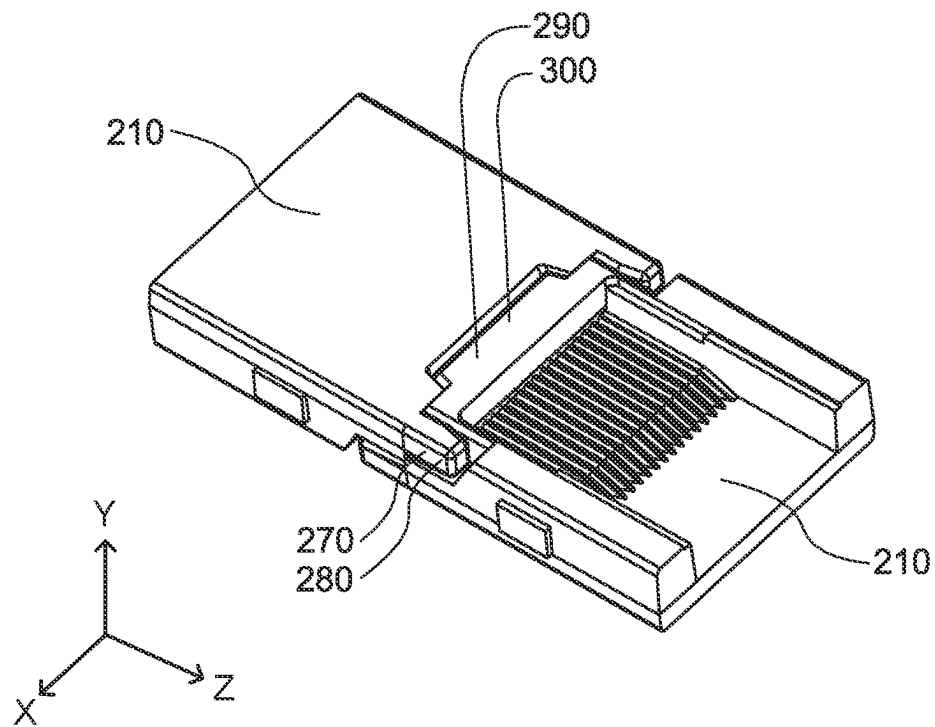


FIG. 5

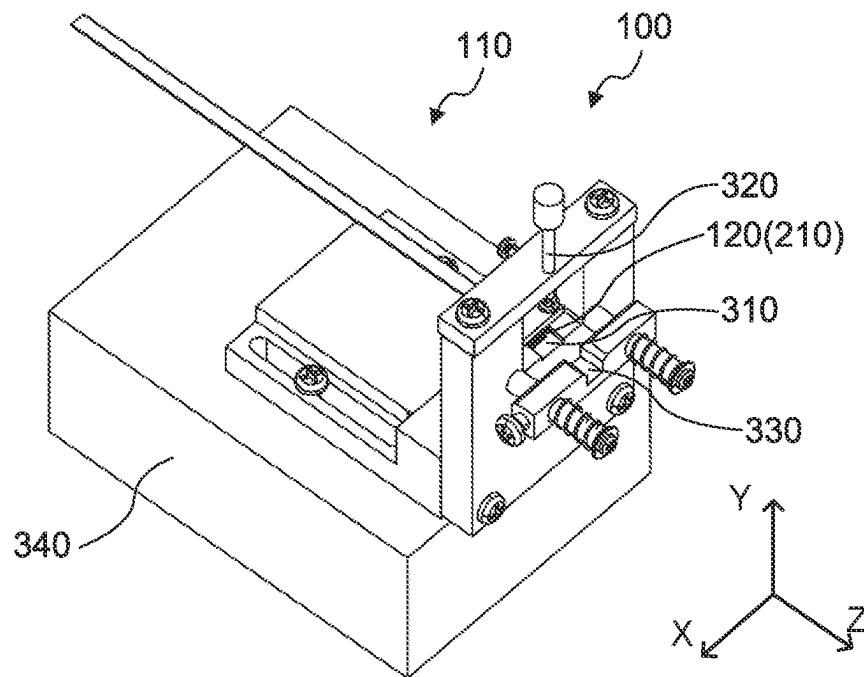


FIG. 6

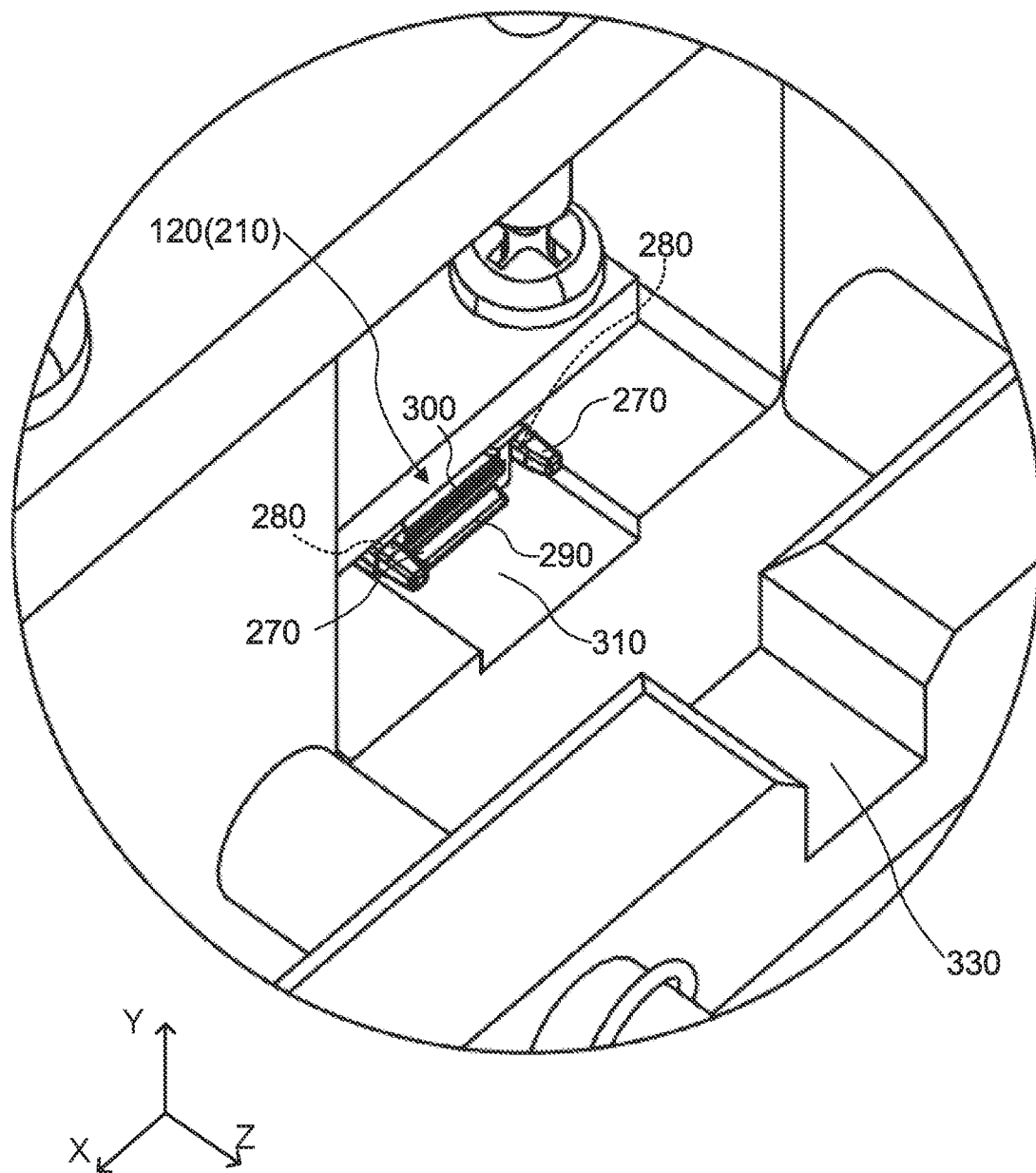


FIG. 7

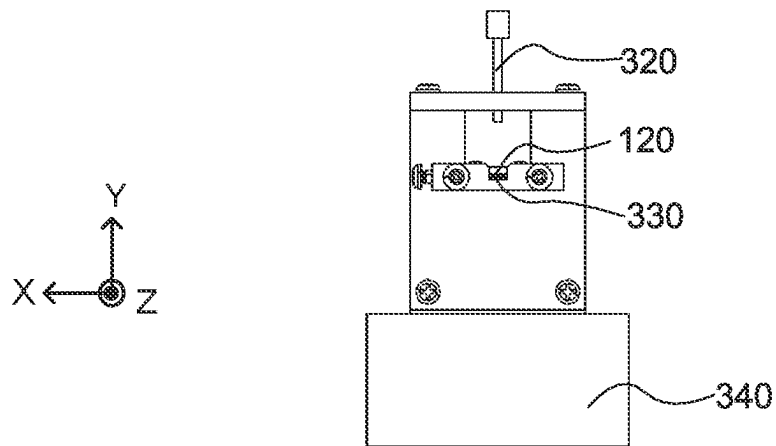


FIG. 8A

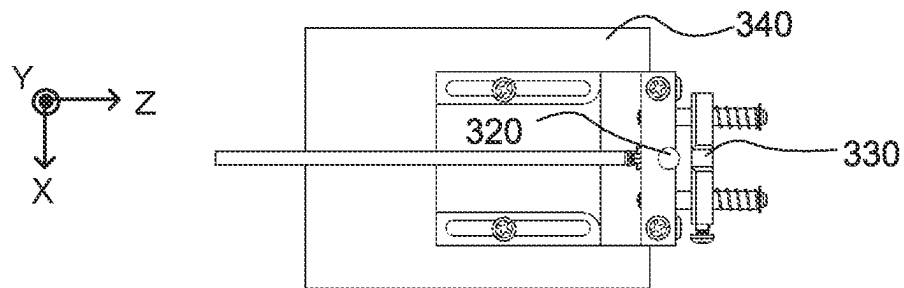


FIG. 8B

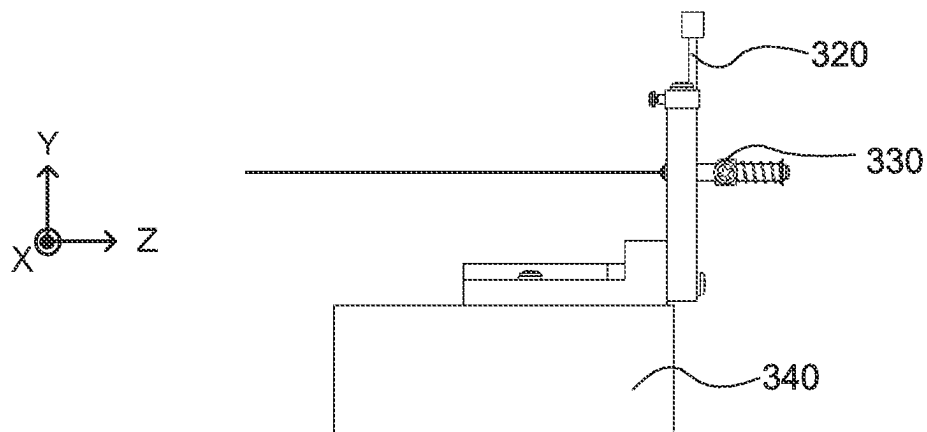
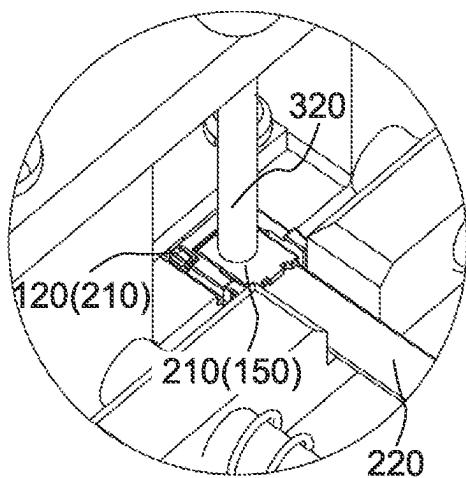
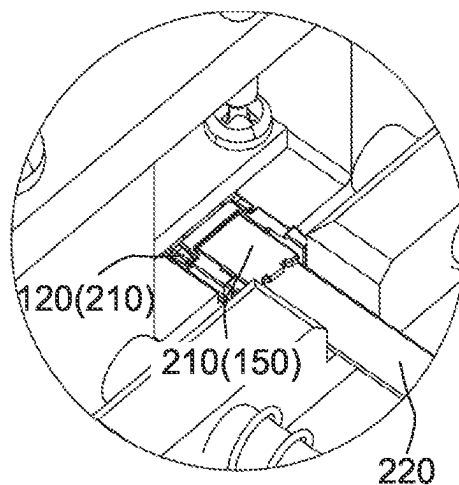
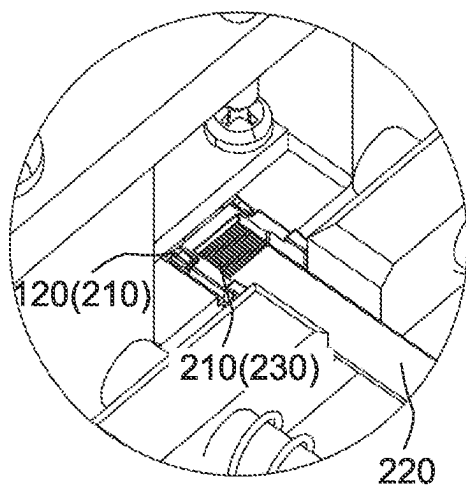
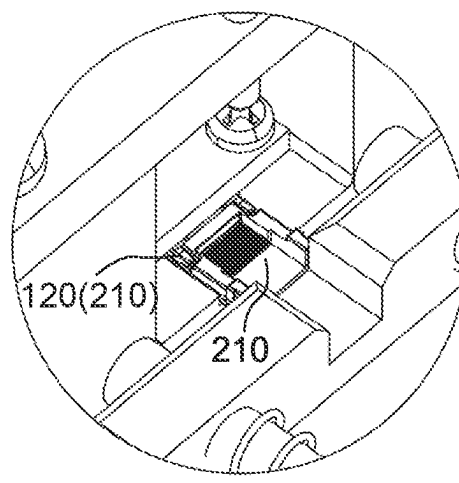
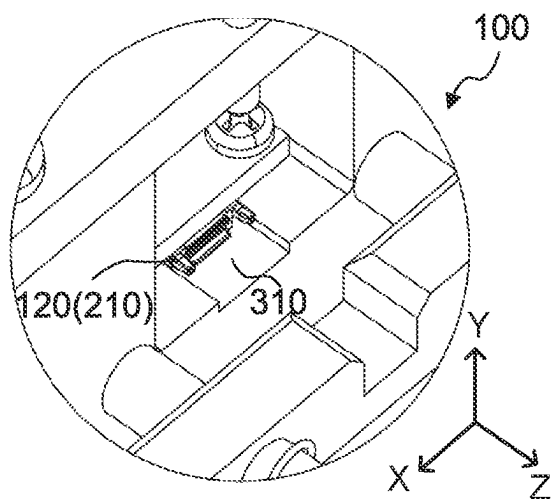


FIG. 8C



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DEVICE FOR PRODUCING OPTICAL CONNECTOR AND METHOD FOR PRODUCING OPTICAL CONNECTOR

CROSS REFERENCE TO RELATED APPLICATIONS

This application is entitled to and claims the benefit of Japanese Patent Application No. 2022-037185, filed on Mar. 10, 2022, the disclosure of which including the specification, drawings and abstract is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a device for producing an optical connector and a method for producing an optical connector.

BACKGROUND ART

A ferrule with an optical transmission member (for example, an optical fiber or an optical waveguide) disposed therein is known. A ferrule with an optical transmission member disposed therein serves as an optical connector and is used to achieve an optical connection.

Patent Literature (hereinafter, referred to as PTL) 1, for example, discloses an optical connector (ferrule) including a substrate and an optical transmission member. The optical connector (ferrule) includes a main body and two positioning pins. The optical connector is used to transmit optical signals to and from the substrate.

CITATION LIST

Patent Literature

PTL 1

Japanese Patent Application Laid-Open No. 2016-180946

SUMMARY OF INVENTION

Technical Problem

For producing an optical connector including a ferrule and an optical transmission member, the ferrule is held by a jig, and the optical transmission member is appropriately disposed relative to the held ferrule. However, ferrules have become smaller in recent years. It has thus become difficult to hold a ferrule in a jig, and when the ferrule is held forcibly, the ferrule may be dented or scratched.

An object of the present invention is to provide a device for producing an optical connector (herein also referred to as "optical connector production device") capable of easily holding a ferrule and reducing the possibility of a dent or a scratch on the ferrule. Another object of the present invention is to provide a method which is for producing an optical connector and which uses the optical connector production device.

Solution to Problem

An optical connector production device of the present invention is configured to produce a first optical connector that includes a first ferrule and a first optical transmission member, the production device including: a jig body; and a

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fixing part for allowing the first ferrule to fit thereto to hold the first ferrule, the fixing part being disposed in the jig body.

A method for producing an optical connector of the present invention uses the production device described above, the method including: holding the ferrule in the fixing part; and disposing one end of the first optical transmission member on the ferrule held by the fixing part.

Advantageous Effects of Invention

The present invention can produce an optical connector without denting or scratching a ferrule even when the ferrule is small. In addition, the present invention can evaluate the positional accuracy of the end of a first optical transmission member during the production of the optical connector.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1A is a perspective view illustrating an optical connector being produced by using a production device according to an embodiment of the present invention, and FIG. 1B is an enlarged view of part of FIG. 1A;

FIG. 2A is a cross-sectional view of the optical connector, and FIG. 2B is a perspective view of a ferrule;

FIG. 3A is a plan view of the ferrule, and FIG. 3B is a bottom surface view of the ferrule;

FIG. 4A is a front view of the ferrule, FIG. 4B is a rear view of the ferrule, FIG. 4C is a left side view of the ferrule, and FIG. 4D is a right side view of the ferrule;

FIG. 5 illustrates a state in which the ferrules are coupled by fitting;

FIG. 6 is a perspective view of the production device;

FIG. 7 is an enlarged view of part of FIG. 6;

FIG. 8A is a front view of the production device, FIG. 8B is a plan view of the production device, and FIG. 8C is a left side view of the production device; and

FIGS. 9A to 9E illustrate how the optical connector is produced by using the production device.

DESCRIPTION OF EMBODIMENTS

Optical Connector being Produced by Production Device

FIG. 1A illustrates optical connector 200 according to an embodiment of the present invention being produced by using production device 100 that is configured to produce optical connector 200. FIG. 1B is an enlarged view of part of FIG. 1A.

In production device 100 of the present embodiment, first ferrule 210 is held by fitting first ferrule 210 in fixing part 120, and first optical transmission members 220 are disposed on the held first ferrule 210 and fixed, thereby producing optical connector 200, as illustrated in FIGS. 1A and 1B.

FIG. 2A is a cross-sectional view of optical connector 200 taken along line A-A of FIG. 1B. As illustrated in FIG. 2A, first optical transmission members 220 are disposed between optical transmission member holding part 230 and lid 150, and fixed in first ferrule 210. Optical connector 200 is thus produced. As illustrated in FIG. 1A, one of light emitting device 10 and light receiving device 20 is optically connected to the end of second optical transmission members 140 connected to fixing part 120, and the other one of light emitting device 10 and light receiving device 20 is optically connected to the end of the first optical transmission members 220 fixed to first ferrule 210. Using production device 100 thus can evaluate whether or not first optical transmission members 220 are optically connected with second

optical transmission members **140** during the production of optical connector **200**. In other words, evaluating the optical coupling of first optical transmission members **220** can evaluate whether or not the end of first optical transmission members **220** are correctly disposed with respect to first ferrule **210**. The detail of such a configuration will be described below. FIG. 1A illustrates light emitting device **10** and light receiving device **20** separated from production device **100**; however, light emitting device **10** and/or light receiving device **20** may be integral with production device **100**. In other words, production device **100** may include light emitting device **10** and/or light receiving device **20**.

Examples of first optical transmission member **220** and second optical transmission member **140** include optical fibers and optical waveguides. In the present embodiment, first optical transmission member **220** and second optical transmission member **140** are optical fibers. In the present embodiment, a plurality of optical fibers are bundled in a row by a covering part to have a ribbon shape.

Configuration of Ferrule

The following describes a configuration of first ferrule **210** at which first optical transmission members **220** are disposed, thereby serving as optical connector **200**.

FIG. 3A is a plan view of first ferrule **210** at which first optical transmission members **220** are disposed, thereby serving as optical connector **200**. FIG. 3B is a bottom surface view of first ferrule **210**. FIG. 3A illustrates first ferrule **210** without lid **150**. FIG. 4A is a front view of first ferrule **210**, FIG. 4B is a rear view of the ferrule, FIG. 4C is a left side view of the ferrule, and FIG. 4D is a right side view of the ferrule.

As illustrated in FIGS. 3A to 4D, first ferrule **210** has a substantially cuboid shape. In addition, in the present embodiment, first ferrule **210** includes optical transmission member holding part **230**, first optical surface **240**, light transmitting wall **250**, second optical surfaces **260**, engaging protrusions **270**, engaging recesses **280**, protrusion **290**, recess **300**, and lid **150**, as illustrated in FIGS. 3A to 4D. The configurations of these components will be described below.

In the following description, the direction along the bottom surface of first ferrule **210** when first ferrule **210** is viewed from the front and rear is defined as "X direction" as illustrated in FIGS. 4A and 4B. A direction orthogonal to the X direction is defined as "Y direction." "Y direction" is the direction (height direction) along the side surface when first ferrule **210** is viewed from the front and from the rear. "Z direction" is orthogonal to "X direction" and "Y direction." In other words, the direction along the bottom surface of first ferrule **210** when first ferrule **210** is viewed from the side is "Z direction," as illustrated in FIGS. 4C and 4D.

Optical transmission member holding part **230** is a portion that holds first optical transmission members **220**. Optical transmission member holding part **230** may have any configuration as long as the holding part can hold first optical transmission members **220**. In the present embodiment, optical transmission member holding part **230** includes grooves **231** where first optical transmission members **220** are disposed. Groove **231** extends perpendicularly to first optical surface **240** in plan view, as illustrated in FIG. 3A. Examples of grooves **231** include V-grooves and U-grooves. Grooves **231** are configured in such a way that when first optical transmission members **220** are disposed, the end surfaces of first optical transmission members **220** are appropriately disposed with respect to first optical surface **240**. The number of grooves **231** may be appropriately set

according to the number of first optical transmission members **220**. In the present embodiment, the number of the grooves is 16.

Optical transmission member holding part **230** may include insertion holes, into which optical fibers are inserted, in place of the grooves. In an optical transmission member holding part including insertion holes, the insertion holes are configured in such a way that when first optical transmission members **220** are inserted, the end surfaces of first optical transmission members **220** are appropriately disposed with respect to first optical surface **240**.

First optical surface **240** directly faces the end surfaces of first optical transmission members **220**. First optical surface **240** has the following configuration: light from first optical transmission members **220** is incident on first optical surface **240**, or light from second optical surfaces **260** is incident on and emitted from first optical surface **240** toward first optical transmission members **220**. First optical surface **240** is disposed on light transmitting wall **250** at the surface opposite to second optical surfaces **260**. In the present embodiment, the first optical surface is parallel to the XY plane.

Light transmitting wall **250** includes first optical surface **240** at one surface and second optical surfaces **260** at the other surface. Light transmitting wall **250** allows transmission of light from first optical surface **240** or second optical surfaces **260** incident on light transmitting wall **250**.

Second optical surface **260** faces first optical surface **240**. Second optical surface **260** has the following configuration: light from first optical surface **240** is incident on and emitted from second optical surface **260**, or light emitted from another optical connector is incident on second optical surface **260**. Second optical surface **260** may be a flat surface or a curved surface. In the present embodiment, second optical surface **260** is a curved surface, and more specifically, a convex lens. The number of second optical surfaces **260** may be appropriately set according to the number of optical transmission members **220** (optical fibers), and may be one or more than one. In the present embodiment, the number of second optical surfaces **260**, which are convex lenses, is more than one in accordance with the number (that is more than one) of optical fibers.

Engaging protrusions **270** are a pair of members projecting in the Z direction and separated from each other to the left and right (separating in the X direction) when first ferrule **210** is viewed from the front, as illustrated in FIG. 4A. In addition, in the present embodiment, engaging protrusion **270** is disposed on the lower side (back surface side) when first ferrule **210** is viewed from the front. In the present embodiment, engaging protrusion **270** has a substantially cuboid shape elongated in the Z direction. Engaging protrusion **270** has a shape complementary to that of engaging recess **280**, and the protrusion and the recess are configured to be able to engage (fit) with each other. In other words, engaging protrusion **270** is a portion that fits into engaging recess **280**.

Specifically, engaging protrusion **270** includes inward restricting surface **271** in the present embodiment. Inward restricting surface **271** faces the inside of first ferrule **210** when first ferrule **210** is viewed from the front. Inward restricting surfaces **271** of the respective paired engaging protrusions **270** face each other in the X direction. This inward restricting surface **271** engages (fits) with outward restricting surface **281** of engaging recess **280**. By engaging (fitting) the surfaces with each other, the movement of first

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ferrule **210** in the left-right direction (X direction) is restricted, thereby fixing the ferrule with respect to the left-right direction.

Engaging recesses **280** are portions recessed in the Z direction and separated from each other to the left and right (separating in the X direction) when first ferrule **210** is viewed from the front, as illustrated in FIG. 4A. In addition, in the present embodiment, engaging recess **280** is disposed on the upper side (top surface side) when the first ferrule is viewed from the front. In the present embodiment, engaging recess **280** has a substantially cuboid shape elongated in the Z direction. Engaging recess **280** has a shape complementary to that of engaging protrusion **270**, and the recess and the protrusion are configured to be able to engage (fit) with each other. In other words, engaging recess **280** is a portion that fits with engaging protrusion **270**.

Specifically, engaging recess **280** includes outward restricting surface **281** in the present embodiment. Outward restricting surface **281** faces the outside of first ferrule **210** when first ferrule **210** is viewed from the front. This outward restricting surface **281** engages (fits) with inward restricting surface **271** of engaging protrusion **270**. By engaging (fitting) the surfaces with each other, the movement of first ferrule **210** in the left-right direction (X direction) is restricted, thereby fixing the ferrule with respect to the left-right direction.

Protrusion **290** is a portion that is disposed on the upper side (top surface side) when first ferrule **210** is viewed from the front, and protrudes in the Z direction. In the present embodiment, protrusion **290** has a substantially cuboid shape elongated in the X direction. Protrusion **290** has a shape complementary to that of recess **300**. Protrusion **290** thus engages with (fits into) recess **300**.

Recess **300** is a portion disposed on the lower side (back surface side) when first ferrule **210** is viewed from the front, and recessed in the Z direction. In the present embodiment, the recess has a substantially cuboid shape elongated in the X direction. Recess **300** has a shape complementary to that of protrusion **290**. Recess **300** thus engages (fits) with protrusion **290**. The engaged (fitted) protrusion **290** and recess **300** fix first ferrules **210** to each other with respect to the Y direction.

As described above, engaging protrusion **270** and engaging recess **280** are engaged (fitted) together, and protrusion **290** and recess **300** are engaged (fitted) together. Specifically, the engagement (fitting) is performed when the front surfaces of two ferrules **210** are put together while the top surface of one of the first ferrules faces up and the back surface of the other one of the first ferrules faces up, as illustrated in FIG. 5.

How first ferrules **210** are fitted together is not limited to the above description. Any manner may be used as long as first ferrules **210** can be fitted together. For example, one first ferrule may have a pin and the other first ferrule may have a hole for receiving the pin, thereby fitting the ferrules together.

Lid **150** is disposed on first optical transmission members **220** disposed at optical transmission member holding part **230**, and holds first optical transmission members **220**. Lid **150** is preferably fixed to first ferrule **210** with an adhesive. The type of adhesive is not limited. Examples of the adhesive include UV curable adhesives.

In the present invention, the first ferrule is not limited to first ferrule **210** described above, and the present invention is also applicable to ferrules having other configurations.

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Production Device

In the following, production device **100** for producing optical connector **200** will be described.

FIG. 6 is a perspective view of production device **100**, and FIG. 7 is an enlarged view of part of FIG. 6. FIG. 8A is a front view of production device **100**, FIG. 8B is a plan view of the production device, and FIG. 8C is a left side view of the production device.

As illustrated in FIG. 6, production device **100** includes jig body **110** and fixing part **120**.

Fixing part **120** is a member for fitting first ferrule **210**. Fixing part **120** is disposed in jig body **110**. Fixing part **120** may be integrated with or separate from production device **100**. Fixing part **120** may have any configuration as long as the fixing part is configured to hold first ferrule **210** by fitting. Fixing part **120** preferably holds first ferrule **210** by fitting with a fitting structure of the first ferrule. The fitting structure is configured to be used when first ferrule **210** is connected to another device (another ferrule). It is preferable that fixing part **120** can hold the front surface of first ferrule **210**. Such a configuration enables fixing of an optical transmission member to first ferrule **210** by irradiating the top surface side or the back surface side of the first ferrule with UV light to cure the UV curable adhesive. The detail of such a configuration will be described below. In addition, the fitting structure of fixing part **120** can preferably be fitted so as to restrict the movement of first ferrule **210** in the XY directions.

In the present embodiment, fixing part **120** includes the following components which enable the fixing part to fit with first ferrule **210**: engaging protrusions **270**, engaging recesses **280**, protrusion **290**, and recess **300**. In the present embodiment, first ferrule **210** itself is used as fixing part **120** having such a configuration (hereinafter, the first ferrule used as the fixing part is also referred to as "second ferrule"), as illustrated in FIG. 7. First ferrule **210** has a structure for fitting with another first ferrules **210** as illustrated in FIG. 5. First ferrule **210** thus can be used as fixing part **120**. Therefore, first ferrule **210**, where first optical transmission members **220** are disposed, can be easily and reliably held in production device **100**, and the possibility of a dent or the like on first ferrule **210** can be reduced.

Jig body **110** may have any configuration as long as the jig body is configured to hold fixing part **120**. In the present embodiment, jig body **110** includes base **340**, ferrule supporting part **310**, lid pressing member **320**, and ferrule pressing member **330**.

Base **340** has a substantially cuboid shape, as illustrated in FIGS. 6 and 8A to 8C. Fixing part **120**, ferrule supporting part **310**, lid pressing member **320**, and ferrule pressing member **330** are disposed on or above base **340**. Base **340** may be mounted in such a way that the base can move in the X, Y, and/or Z direction and/or rotatably mounted.

Ferrule supporting part **310** is a portion for supporting first ferrule **210** fitted to fixing part **120**. More specifically, ferrule supporting part **310** supports first ferrule **210** from below so as to prevent first ferrule **210** held by fixing part **120** from moving downward and coming off fixing part **120** when first ferrule **210** is pushed from above by lid pressing member **320**. In the present embodiment, ferrule supporting part **310** is a recess. The recess preferably has a Z-direction length that is equal to or greater than the Z-direction length of lid **150** of first ferrule **210** or equal to or greater than the Z-direction length of optical transmission member holding part **230** of first ferrule **210**. A recess having such a length enables reliable pressing of lid **150** toward the first optical transmission member **220** side when first optical transmission members are disposed at first ferrule **210** held by fixing

part **120** and then lid **150** is disposed thereon (see the cross-sectional view of FIG. 2A).

Lid pressing member **320** is a portion for pressing lid **150** (see FIG. 2A) of first ferrule **210** held by fixing part **120**. Lid pressing member **320** may be disposed at any position above ferrule supporting part **310** as long as the member can press lid **150** of held first ferrule **210**. In the present embodiment, lid pressing member **320** includes a rod-shaped member elongated in the Y direction. In addition, lid pressing member **320** is configured to be movable in the Y direction in the present embodiment. Any method may be used for configuring the lid pressing member to be movable in the Y direction. In the present embodiment, the surface of the rod-shaped member of lid pressing member **320** has threads, and rotating the rod-shaped member can move the member in the Y direction to press lid **150**.

Ferrule pressing member **330** presses first ferrule **210** fitted to fixing part **120** toward fixing part **120** (in the Z direction). Further, the ferrule pressing member is a portion that supports first optical transmission members **220** disposed at ferrule **210**. In the present embodiment, ferrule pressing member **330** includes a recess that supports the ribbon-shaped first optical transmission member **220**. In addition, ferrule pressing member **330** is configured to be movable in the Z direction in the present embodiment. Method for Producing Optical Connector

FIGS. 9A to 9E illustrate a method for producing optical connector **200**. In the following, a method for producing optical connector **200** will be described with reference to the drawings.

As illustrated in FIG. 9A, production device **100** including fixing part **120** (second ferrule **210**) is prepared. As illustrated in FIG. 9B, first ferrule **210** with lid **150** removed is disposed on ferrule supporting part **310**. First ferrule **210** is fixed to fixing part **120** by applying a pressing force to the first ferrule with ferrule pressing member **330**.

As illustrated in FIG. 9C, the end of first optical transmission members **220** is disposed on first ferrule **210** held by fixing part **120**. An adhesive (for example, a UV curable adhesive) is then applied from above optical transmission member holding part **230** of first ferrule **210**.

As illustrated in FIG. 9D, lid **150** is placed on the end of first optical transmission members **220** disposed on first ferrule **210**. As illustrated in FIG. 9E, lid pressing member **320** then presses lid **150**.

The present embodiment describes the following (see FIG. 1A): one of light emitting device **10** and light receiving device **20** is disposed at the other end of first optical transmission members **220** whose one end is disposed at first ferrule **210**; and the other one of light emitting device **10** and light receiving device **20** is disposed at the other end of second optical transmission members **140** whose one end is disposed at fixing part **120** (second ferrule **210**). Such a configuration allows evaluation of whether or not first optical transmission members **220** are optically and properly coupled with second optical transmission members **220** by emitting light from light emitting device **10** and detecting the light transmitted through the first optical transmission members and the second optical transmission members, by using light receiving device **20**. Specifically, evaluating whether or not the optical transmission members are optically and properly coupled can evaluate the position of the end of first optical transmission members **220** in first ferrule **210**.

When the position of the end of first optical transmission members **220** is evaluated as appropriate, first optical transmission members **220** can be fixed to first ferrule **210** by, for

example, irradiating the UV curable adhesive with UV from above to cure the adhesive. In the present embodiment, first ferrule **210** is held in fixing part **120** by fitting the fitting structure disposed at the front surface of the first ferrule with the fixing part. As a result, nothing that would block the UV is on the top surface or the back surface of first ferrule **210**, and the UV curable adhesive can be easily cured.

On the other hand, when the position of the end of first optical transmission members **220** is evaluated as inappropriate, first optical transmission members **220** can be reattached and the position of the end of first optical transmission members **220** can be evaluated again.

Effect

According to the optical connector production device of the present invention and the method which is for producing an optical connector and which uses the optical connector production device, first ferrule **210** where first optical transmission member **220** is disposed can be held by fitting. First ferrule **210** thus can be easily and reliably held without, for example, denting first ferrule **210**. In the present embodiment, first ferrule **210** or a component having a similar structure serves as fixing part **120**, and second optical transmission member transmission member **140** is connected to fixing part **120**. The position of the end of first optical transmission member **220** in first ferrule **210** thus can be evaluated with first optical transmission member **220** disposed in first ferrule **210**. Therefore, whether or not to fix first optical transmission member **220** to first ferrule **210** with an adhesive or the like can be decided after the evaluation. The yield of producing the optical connector thus can be improved.

INDUSTRIAL APPLICABILITY

The optical connector production device and the method for producing an optical connector according to the present invention are particularly advantageous for producing optical connectors including ferrules and optical transmission members.

REFERENCE SIGNS LIST

- 10** Light emitting device
- 20** Light receiving device
- 100** Production device
- 120** Fixing part
- 140** Second optical transmission member
- 150** Lid
- 200** Optical connector
- 210** First ferrule (second ferrule)
- 220** First optical transmission member
- 230** Optical transmission member holding part
- 231** Groove
- 240** First optical surface
- 250** Light transmitting wall
- 260** Second optical surface
- 270** Engaging protrusion
- 271** Inward restricting surface
- 280** Engaging recess
- 281** Outward restricting surface
- 290** Protrusion
- 300** Recess
- 310** Ferrule supporting part
- 320** Lid pressing member
- 330** Ferrule pressing member
- 340** Base

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The invention claimed is:

1. A method for producing an optical connector, using an optical connector production device, wherein

the optical connector includes a first ferrule and a first optical transmission member

the optical connector production device comprises:

a jig body; and

a fixing part for allowing the first ferrule to fit thereto to hold the first ferrule, the fixing part being disposed in the jig body,

the method comprises:

fitting the first ferrule to the fixing part; and

disposing one end of the first optical transmission member on the first ferrule fitted to the fixing part.

2. The method according to claim 1, wherein

the optical connector production device further comprises a light emitting device and a light receiving device for evaluating optical coupling of the first optical transmission member disposed at the first ferrule,

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the method further comprises:

optically connecting one of the light emitting device or the light receiving device to another end of the first optical transmission member including the one end disposed on the first ferrule; and

evaluating positional accuracy of the one end of the first optical transmission member in the first ferrule by detecting light emitted from the light emitting device and transmitted through the first optical transmission member and a second optical transmission member, the detecting being performed by using the light receiving device.

3. The method according to claim 2, further comprising, after the evaluating, fixing the one end of the first optical transmission member to the first ferrule.

4. The method according to claim 3, wherein the fixing includes curing a UV curable adhesive by UV irradiation, the UV curable adhesive being previously adhered to the one end of the first optical transmission member, the one end being held by the first ferrule.

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